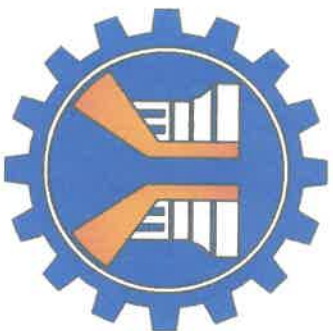




Republic of the Philippines
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
LA UNION SECOND DISTRICT ENGINEERING OFFICE
San Joaquin Sur, Agoo, La Union, Region I



BAGONG PILIPINAS

C.Y. 2024 PROJECT
DETAILED ENGINEERING DESIGN PLAN FOR
**SIPAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR / STRATEGIC PUBLIC
BUILDINGS / FACILITIES - CONSTRUCTION OF SHORE PROTECTION WORKS,
BARANGAY URAYONG, BAUANG, LA UNION**

SECTION: BARANGAY URAYONG, BAUANG, LA UNION
LOCATION: BARANGAY URAYONG, BAUANG, LA UNION
STATION LIMITS: STA. 0+000 - STA. 0+500
NET LENGTH: 275.00 LINEAL METERS

SUBMITTED:


JOSE D. PEREZ
CHIEF, PLANNING AND DESIGN SECTION

DATE:

RECOMMENDED:

RAUL P. GALI
ASSISTANT DISTRICT ENGINEER

DATE:

APPROVED:

GIL B. LORENZO
DISTRICT ENGINEER

DATE:

HYDRAULIC GEOTEXTILE TUBE ITEM 715(8)

GEOTEXTILE TUBE IS MANUFACTURED USING SPECIALLY ENGINEERED WOVEN FABRICS TO PROVIDE EXCELLENT ROBUSTNESS, ABRASION RESISTANCE, UV PROTECTION AND DURABILITY IN MARINE AND HYDRAULICS APPLICATION.

MECHANICAL PROPERTIES	TEST STANDARD	UNIT	MINIMUM AVERAGE VALUES	
			Machine Dr.	Core Dr.
PROPERTY	100% UV STABILIZED POLYPROPYLENE REINFORCED HIGH STRENGTH REINFORCEMENT FABRIC			
WIDE WIDTH TENSILE STRENGTH (AT ULTIMATE)	ISO 10319	kN/m	120 (min.)	120 (min.)
STRAIN AT NOMINAL TENSILE STRENGTH	ISO 10319	%	10 (max.)	10 (max.)
CRR PUNCTURE STRENGTH	ISO 12235	kN	14 (min.)	
FACTORY BEAM STRENGTH	ISO 10321	kN/m	80 (min.)	
DROP CONE PERFORATION	ISO 13433	mm	12 (max)	
DIAMETER				
PORE SIZE	ISO 12956	mm	0.4	
WATER PERMEABILITY	ISO 11059	l/m ² /s	15 (min.)	
ABRASION RESISTANCE	ASTM D4958	Washload	75 (min.)	
UV RESISTANCE (AT 500 HOURS)	ASTM D4355	%retained	90 (min.)	

SEPARATION GEOTEXTILE ITEM 715(2)

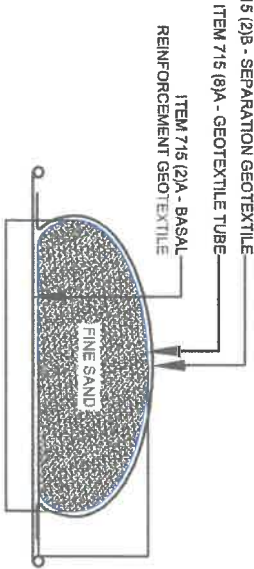
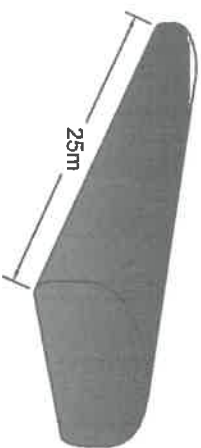
THE BASAL REINFORCEMENT GEOTEXTILE IS MADE FROM HIGHLY SPECIALIZED FIBER TECHNOLOGY GEOTEXTILE THAT PROVIDES EXCELLENT ABRASION RESISTANCE AND DURABILITY OVER TIME.

MECHANICAL PROPERTIES	TEST STANDARD	UNIT	MINIMUM AVERAGE VALUES	
			Machine Dr.	Core Dr.
PROPERTY	100% UV STABILIZED POLYPROPYLENE REINFORCED HIGH STRENGTH REINFORCEMENT FABRIC			
TENSILE STRENGTH	ISO 10319	kN/m	125 (min.)	125 (min.)
STRAIN AT NOMINAL TENSILE STRENGTH	ISO 10319	%	15 (max.)	15 (max.)
CRR PUNCTURE STRENGTH	ISO 12235	kN	14 (min.)	
DROP CONE PERFORATION	ISO 13433	mm	12 (max)	
DIAMETER	ISO 12956	mm	0.35 (max)	
PORE SIZE	ISO 11059	l/m ² /s	15 (min.)	
WATER PERMEABILITY	ASTM 4586	%retained	75 (min.)	
ABRASION RESISTANCE (AT 500 HOURS)	ASTM D4355	%retained	90 (min)	
LOOP SPACING (IN LOOP PROPERTIES)		m	0.25 (min)	1 (min)
LOOP TENSILE STRENGTH		kN	2.0 (min)	

DRAINAGE GEOTEXTILE ITEM 715(3)

TWO LAYER MECHANICALLY BONDED POLYPROPYLENE CONTINUOUS FILAMENT NONWOVEN NEEDLE PUNCHED 100% POLYPROPYLENE, UV STABILIZED

MECHANICAL PROPERTIES	TEST STANDARD	UNIT	MINIMUM AVERAGE VALUES
TENSILE STRENGTH MD/CD	ISO 10319	kN/m	3000 (min.)
TENSILE ELONGATION MD/CD	ISO 10319	%	30/40 (max.)
CRR PUNCTURE STRENGTH	ISO 12235	N	4500 (min.)
EFFECTIVE OPENING SIZE C_{90}	ISO 12958	mm	0.08 (max.)
THICKNESS 2 σ_{10}	ISO 9903	mm	4.50 (min.)
NOMINAL MASS	ISO 9884	g/m ²	600 (min.)
PERMEABILITY NORMAL TO THE PLANE (mm H ₂ O)	ISO 11053	l/m ² /s	35
CHEMICAL / BIOLOGICAL RESISTANCE	RESISTANT AGAINST ALL CHEMICAL AGENTS AND MICROORGANISMS USUALLY OCCURRING IN SEAS OR RIVERS		



HYDRAULIC GEOTEXTILE TUBE

METHODS OF STORAGE

THE GEOTEXTILES DELIVERED TO THE SITE SHALL BE STORED IN COVERED AREA AND LEVEL FOUNDATION. THE GEOTEXTILE SHALL BE PREFERABLY BE PLACED ON WOODEN BEAM SUPPORT (ELEVATD 50mm ABOVE GROUND LEVEL) TO AVOID DIRECT CONTACT WITH THE GROUND LEVEL. THIS IS TO PREVENT THE GEOTEXTILE FROM GETTING SOAK OR WET DURING RAIN FLOODING.

THE GEOTEXTILE SHALL BE STACKED IN TRIANGULAR MANNER NOT MORE THAN 10 ROLLS IN HEIGHT AND BE COVERED WITH TARPULIN OR PLASTIC SHEET AND FINALLY WEIGHTED DOWN. WHERE PROPER COVERED STORAGE ARE IS NOT AVAILABLE THE ENTIRE STACKED STACKED ROLLS OF GEOTEXTILES SHALL BE COVERED WITH TARPULIN OR PLASTIC SHEET AND FINALLY WEIGHTED DOWN.

SUPPLY OF SAND AND WATER

GEOTUBES ARE FILLED BY HYDRAULICALLY PUMPING FILL MATERIAL INTO THE TUBE. TO ALLOW FLOW OF FILL MATERIAL, THE FILL IS IN THE FORM OF SAND/WATER SLURRY. THE GEOTUBE BEING PERMEABLE, ALONGS THE EXCESS WATER TO PASS THROUGH THE GEOTEXTILE SKIN WHILE SAND IS REMAINING INSIDE THE TUBE. ONLY SAND MATERIAL FROM APPROVED SOURCES AND MEETING SPECIFIED ENGINEERING CRITERIA SHOULD BE USED. SUPPLY OF WATER IS USUALLY FREE AND ABUNDANT AT SITE AS SEA WATER MAY BE USED TO CREATE THE SAND SLURRY.

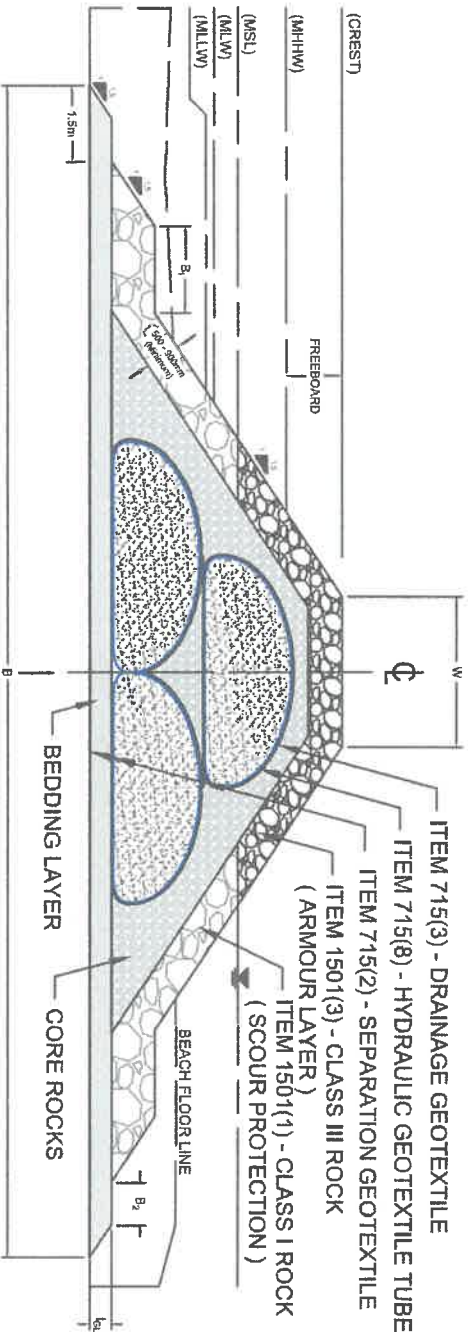
THE GRADING OF THE SELECTED SAND FILL SHOULD BE IN ACCORDANCE WITH THE GRADING REQUIREMENT SPECIFIED IN TABLE. FINER GRADATION LIMITS OF SAND FILL MAY BE PERMISSIBLE SUBJECT TO A FILTRATION REQUIREMENT CHECK AND THE APPROVAL OF THE ENGINEER.

15 (max.) MECHANICAL FILLING PERCENTAGE PASSING (%)		MECHANICAL FILLING PERCENTAGE PASSING (%)	
125 (min.)	100	100	
125 (min.)	70-100	30-100	
125 (min.)	20-60	5-40	
125 (min.)	0-20	0-10	

TABLE 1a: GRADING SPECIFICATION FOR SAND FILL

DESCRIPTION	DIAMETER	TIME (SETTLING)	
		mm	SECOND
GRAVEL	10.0	1.0	
COARSE SAND	1.00	10	
FINE SAND	0.10	125	
SILT	0.075	108	

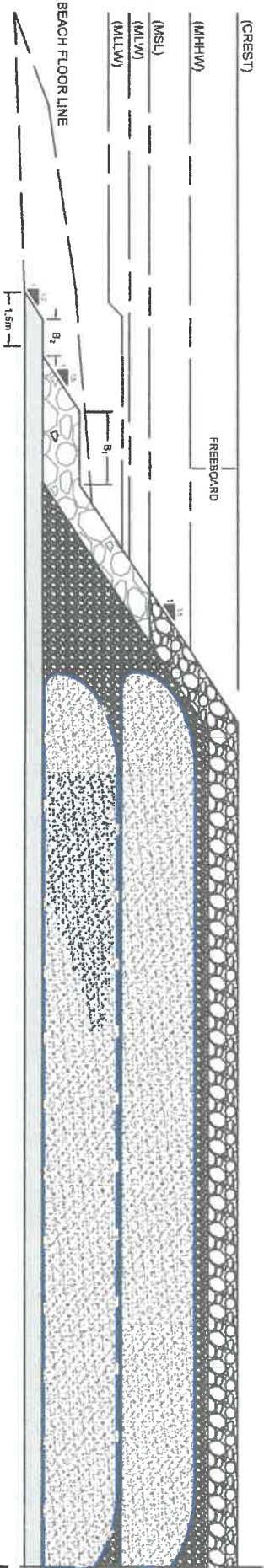
LAYER	MINIMUM ROCK WEIGHT	
	STRUCTURE HEAD	STRUCTURE TRUNK
ARMOUR LAYER	500 kg/pc	750 kg/pc
CORE ROCK	3,75 kg/pc	3,75 kg/pc
SCOUR PROTECTION	2000 kg/pc	3000 kg/pc



TYPICAL SECTION

SCALE

NTS

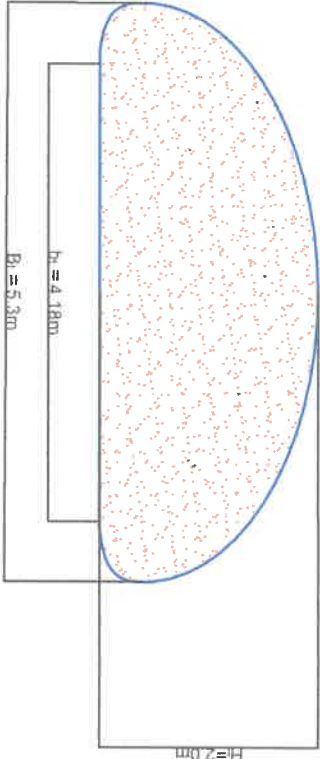


LONGITUDINAL SECTION

SCALE

NTS

A. Against Wave Attack



Notes:

Computation for External Stability

- H_t =Filled Height of Geotube
- B_t =Filled Width of Geotube
- b_t =Base Contant Width of Geotube

From: Geosystems Design Rules and Applications (A. Bezuijen & E.W. Vastenburg, 2013)

TenCate Design Analysis

Design Calculation Against Wave Attack
External Stability

$$\frac{H_s}{(\Delta t)(D_k)} \leq 1.0$$

- H_s = significant Wave height (m)
- (Δ_t) = relative density of the geotextile tube
- D_k = Efective thickness of the geotextile tube
- B_t = width of geotextile tube
- L_t = length of geotextile tube

Computation Against Wave Attack

- Significant wave height (H_s) conservatively assumed to be 2.0m.

- H_s = 2.0m
- (Δ_t) = 1
- B_t = 5.3m
- D_k = 2(5.3m) = 10.6m (Parallel to direction of wave attack)
- D_k =5.3m (Perpendicular to direction of wave attack)

Significant wave height (H_s) conservatively assumed to be 2.0m.

Computation Against Wave Attack

- H_s = 2.0m
- (Δ_t) = 1
- B_t = 5.3m
- D_k = 2(5.3m) = 10.6m (Parallel to direction of wave attack)
- D_k =5.3m (Perpendicular to direction of wave attack)

Parallel to direction of wave attack

$$\frac{H_s}{(\Delta t)(D_k)} \leq 1.0$$

$$\frac{H_s}{(\Delta t)(D_k)} = \frac{2.0}{(1)(10.6)} = 0.189$$

0.189 ≤ 1.0 Complies against to parallel wave attack

Perpendicular to direction of wave attack

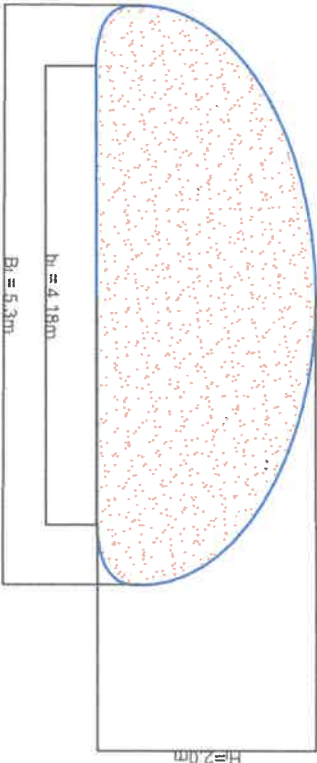
$$\frac{H_s}{(\Delta t)(D_k)} \leq 1.0$$

$$\frac{H_s}{(\Delta t)(D_k)} = \frac{2.0}{(1)(5.3)} = 0.377$$

0.377 ≤ 1.0 Complies against to perpendicular wave attack

 Republic of the Philippines DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS LA UNION SECOND DISTRICT ENGINEERING OFFICE San Joaquin St., Agoo, La Union, Region I BACODANG PILIPINAS	PROJECT NAME AND LOCATION: SIPAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR /STRATEGIC PUBLIC BUILDINGS / FACILITIES - CONSTRUCTION OF SHORE PROTECTION WORKS BARANGAY URAYONG, BAYANG, LA UNION BARANGAY URAYONG, BAYANG, LA UNION	SHEET CONTENTS: GENERAL NOTES AND SPECIFICATIONS	DRAFTER: RODOLFO M. GONG ENGR. ENGR. ASSISTANT PREPARED: RODOLFO M. GONG ENGR. ENGR. II	REVIEWED: JOSE P. MAQUINQUEZ ENGINEER II	SUBMITTED: JOSE P. MAQUINQUEZ CHIEF / PLANNING AND DESIGN SECTION	RECOMMENDED: RAUL P. GALI ASSISTANT DISTRICT ENGINEER	APPROVED: GIL B. LORENZO DISTRICT ENGINEER	SET NO. 	SHEET NO. 03 09
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B. Against Current Attack



From: Geosystems Design Rules and Applications (A. Bezuijen & E.W. Vastenburt, 2013)

TenGate Design Analysis

Design Calculation Against Current Attack
External Stability

$$U_{cr} \leq 1.2 \sqrt{(g)(\Delta_t)(D_k)}$$

- U_{cr} = critical water flow velocity (m/s)
- (Δ_t) = relative density of the geotextile tube
- D_k = Effective thickness of the geotextile tube
- B_t = width of geotextile tube
- L_t = length of geotextile tube
- g = acceleration due to gravity ($g = 9.81 \text{ m/s}^2$)

Computation Against Current Attack

Critical water flow is not given and assumed to be 3.0 m/s, river currents.

$$\begin{aligned} U_{cr} &= 3.0 \text{ m/s} \\ (\Delta_t) &= 1 \\ B_t &= 5.3 \text{ m} \\ D_k &= 2(5.3 \text{ m}) = 10.6 \text{ m (Parallel to direction of current attack)} \\ D_k &= 5.3 \text{ m (Perpendicular to direction of current attack)} \end{aligned}$$

Parallel to direction of current attack

$$U_{cr} = \frac{3.0}{\sqrt{(g)(\Delta_t)(D_k)}} = \frac{3.0}{\sqrt{(9.81)(1)(10.6)}} = 0.294$$

0.343 $\leq$ 1.2 Complies against to parallel current attack

Perpendicular to direction of current attack

$$U_{cr} = \frac{3.0}{\sqrt{(g)(\Delta_t)(D_k)}} = \frac{3.0}{\sqrt{(9.81)(1)(5.3)}} = 0.416$$

0.485 $\leq$ 1.2 Complies against to perpendicular current attack

 Republic of the Philippines DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS LA UNION SECOND DISTRICT ENGINEERING OFFICE San Juanan Sur, Agoo, La Union, Region I  BAGUIO CITY OFFICE		PROJECT NAME AND LOCATION: SIPAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR STRATEGIC PUBLIC BUILDINGS / FACILITIES - CONSTRUCTION OF SHORE PROTECTION WORKS BARANGAY URAYONG, BAWANG, LA UNION		SHEET CONTENTS: GENERAL NOTES AND SPECIFICATIONS		DRAFTED: ROZEL M. LAGUNA ENGINEERING ASSISTANT PREPARED: ROZEL M. LAGUNA ENGINEER II DATE: 10/10/2023		REVIEWED: JOSE P. MORALES ENGINEER II DATE: 10/10/2023		SUBMITTED: JOSE D. PEREZ CHIEF PLANNING AND DESIGN SECTION DATE: 10/10/2023		RECOMMENDED: RAUL P. GALI ASSISTANT DISTRICT ENGINEER DATE: 10/10/2023		APPROVED: GIL B. LORENZO DISTRICT ENGINEER DATE: 10/10/2023		SET NO. 04		SHEET NO. 09	
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LEGEND

SYMBOLS

ABBREVIATION

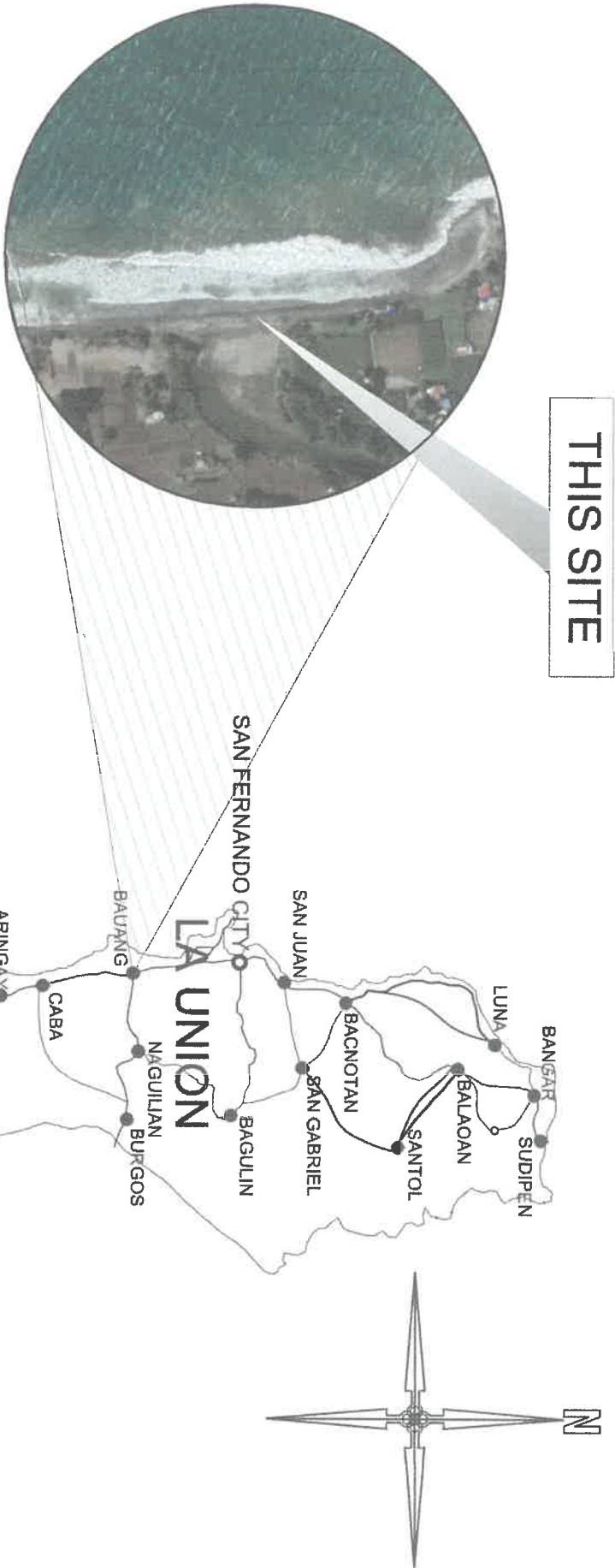
LIMIT OF DIMENSION

BENCH MARK

BOREHOLE

WATER LEVEL

NORTH ARROW



INDEX OF SHEET

SHEET NO.	SHEET CONTENT
1 09	COVER PAGE
2-4 09	GENERAL NOTES AND SPECIFICATIONS
5-6 09	TYPICAL CROSS SECTION
7 09	LEGEND INDEX OF SHEET SUMMARY QUANTITIES VICINITY MAP
8-9 09	PLAN

SUMMARY OF QUANTITIES

ITEM	DESCRIPTION	UNIT	QUANTITIES	REMARKS
1702(1)a	STRUCTURE EXCAVATION (COMMON SOIL)	CU.M	20,118.08	
715(2)	SEPARATION GEOTEXTILE (WOVEN FILTER GEOTEXTILE)	SQ.M.	11,396.00	
715(3)	DRAINAGE GEOTEXTILE	SQ.M.	6,930.00	
715(8)	HYDRAULIC GEOTEXTILE TUBE	PCS.	33.00	
1501(1)a	ROCK WORKS, CLASS I ROCK	TONNE	12,007.85	
1501(1)b	ROCK WORKS, CLASS I ROCK	TONNE	2,301.85	
1501(3)c	ROCK WORKS, CLASS III ROCK	TONNE	6,444.45	
A.1.1(8)	PROVISION OF FIELD OFFICE FOR THE ENGINEER (RENTAL BASIS)	MONTH	8.83	
B.5(1)	PROJECT BILLBOARD/SIGNBOARD	EACH	2.00	
B.7(1)	OCCUPATIONAL SAFETY AND HEALTH PROGRAME	L.S.	1.00	
B.9(1)	MOBILIZATION/DEMOBILIZATION	L.S.	1.00	
B.18(1)	CRANEWAY	L.M.	250.00	

Republic of the Philippines
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
LA UNION SECOND DISTRICT
ENGINEERING OFFICE
San Josean St., Agoo, La Union, Region 1

PROJECT NAME AND LOCATION:
SIYAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR STRATEGIC PUBLIC BUILDINGS, FACTORIES, CONSTRUCTION OF SHORE PROTECTIVE WORKS, BARANAGAY URAYONG, BAUANG, LA UNION
BARANAGAY URAYONG, BAUANG, LA UNION

SHEET CONTENTS:
LEGEND
INDEX OF SHEET
SUMMARY QUANTITIES
VICINITY MAP

DATE: 07/09/2024

ENGINEER II: ROSELYN D. VERA

DATE: 07/09/2024

ENGINEER I: JOSE P. MORALES

DATE: 07/09/2024

CHIEF PLANNING AND DESIGN SECTION: JOSE D. BEREZ

DATE: 07/09/2024

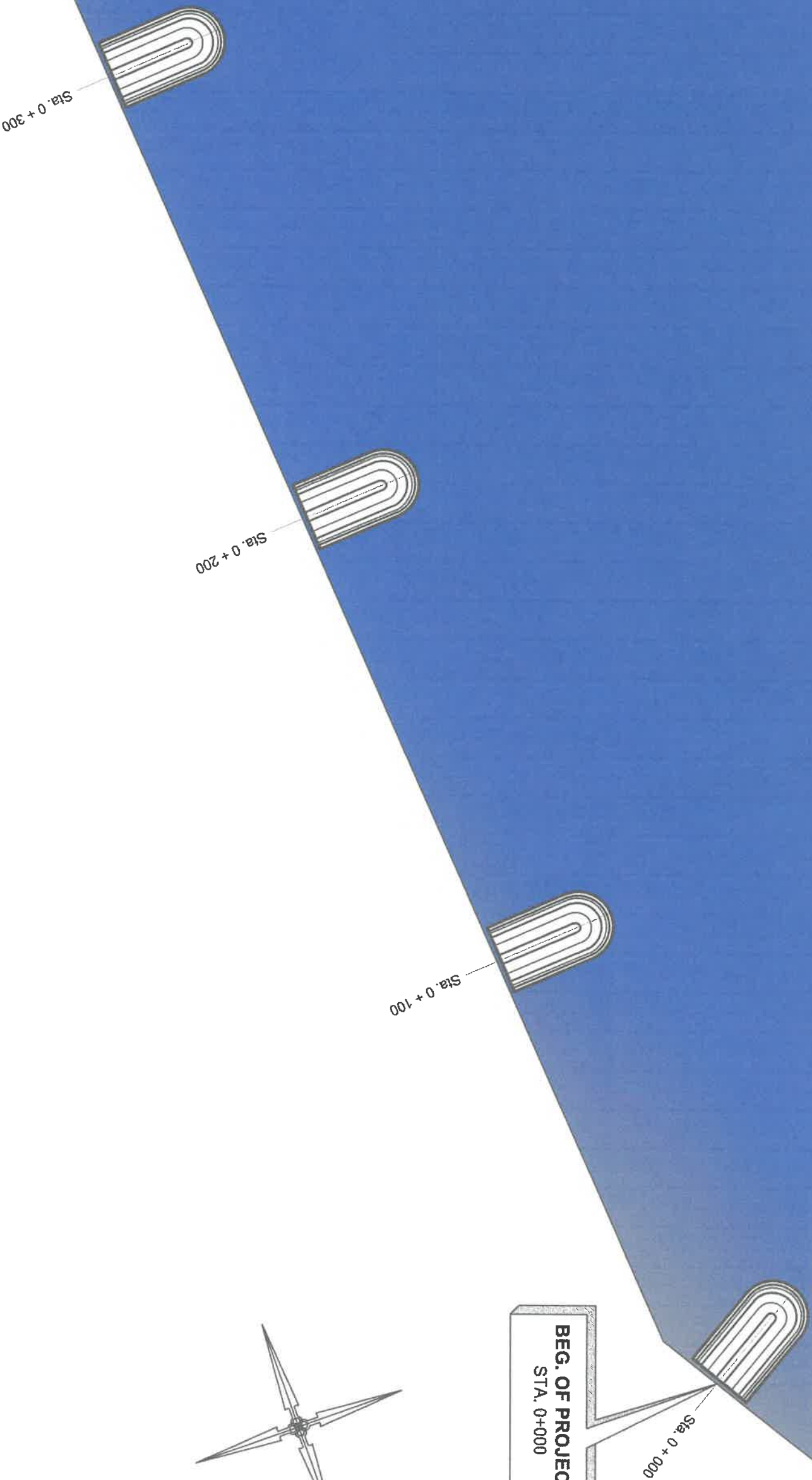
ASSISTANT DISTRICT ENGINEER: RAUL P. GALI

DATE: 07/09/2024

DISTRICT ENGINEER: GIL B. LORENZO

1
09

07
09



LEGEND :



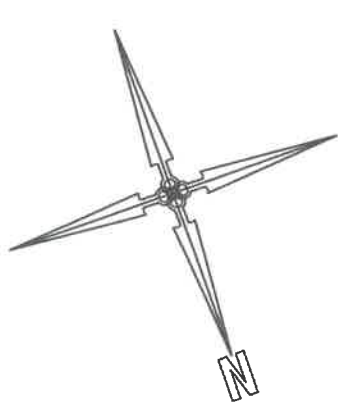
PROPOSED GROINS



SEA

PLAN
SCALE: 1:700 M

Republic of the Philippines DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS LA UNION SECOND DISTRICT ENGINEERING OFFICE San Josem Sur, Agoo, La Union, Region I LAGOON PHILIPPINES		PROJECT NAME AND LOCATION: SIPAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR STRATEGIC PUBLIC BUILDINGS / FACILITIES - CONSTRUCTION OF SHORE PROTECTION WORKS BARANGAY URAYONG, BALANG, LA UNION		SHEET CONTENTS: PLAN		DRAFTED: ROSA M. MENDO ENGINEERING ASSISTANT PREPARED: ROSA M. MENDO ENGINEER I		REVIEWED: JOSE P. MARQUEZ ENGINEER II DATE:		SUBMITTED: JOSE D. PENEZ CHIEF, PLANNING AND DESIGN SECTION DATE:		RECOMMENDED: RAUL P. GALLI ASSISTANT DISTRICT ENGINEER DATE:		APPROVED: GIL B. LORENZO DISTRICT ENGINEER DATE:		SET NO. 	SHEET NO.
--	--	---	--	-----------------------------	--	--	--	--	--	--	--	---	--	---	--	-------------	---------------



PLAN

SCALE: 1:700 M

LEGEND :



PROPOSED GROINS



SEA

 Republic of the Philippines DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS LA UNION SECOND DISTRICT ENGINEERING OFFICE San Josefin Str, Agoo, La Union, Region I  BARANGAY URAYONG, BAYANG, LA UNION		PROJECT NAME AND LOCATION:	SHEET CONTENTS:		DATE:		REVIEWED:		SUBMITTED:		RECOMMENDED:		APPROVED:		SET NO.		SHEET NO.	
SIPAG - FLOOD MITIGATION STRUCTURES PROTECTING MAJOR / STRATEGIC PUBLIC BUILDINGS / FACILITIES - CONSTRUCTION OF SHORE PROTECTION WORKS - BARANGAY URAYONG, BAYANG, LA UNION		PLAN		PREPARED: ROZEL M. DE VERA ENGINEER III		JOSIE P. MARQUEZ ENGINEER II		JOSE B. PEREZ CHIEF, PLANNING AND DESIGN SECTION		RAUL P. GALL ASSISTANT DISTRICT ENGINEER		GIL B. LORENZO DISTRICT ENGINEER						